

UNIVERSITY OF TEHRAN
FACULTY OF ELECTRICAL AND COMPUTER ENGINEERING

Technical Report: Real-Time Biopotential ECG Acquisition, Hardware Front-End Design & Heart Rate Variability (HRV)

Course:	Introduction to Biomedical Engineering (IBME) -- Fall 2023
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1. Abstract & System Architecture

2. Hardware Front-End (AD8232 IC & Pinout)

The AD8232 is an integrated biopotential front-end designed specifically to extract, amplify, and filter small biopotential signals in the presence of noisy conditions, such as motion or remote electrode placement. It features an instrumentation amplifier with an uncommitted operational amplifier for adjustable gain and a right leg drive (RLD) amplifier to maximize Common-Mode Rejection Ratio (CMRR > 80 dB).

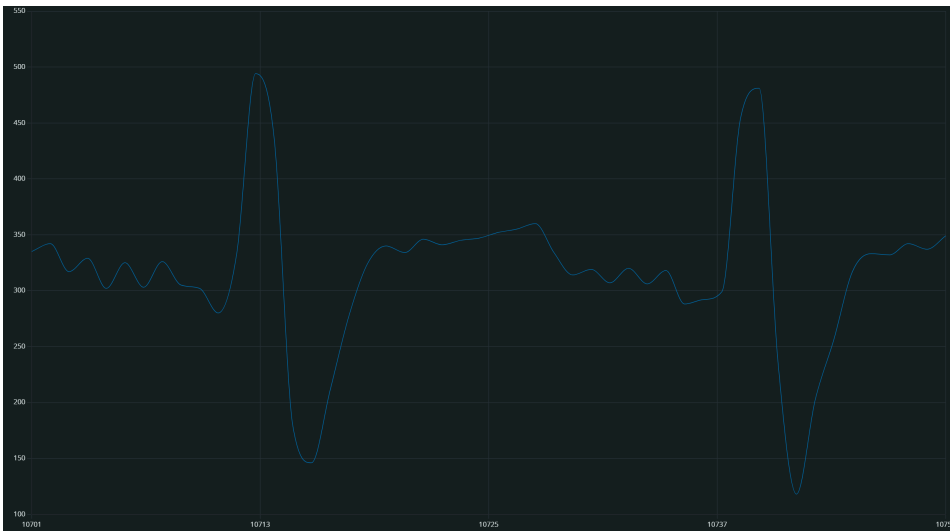


Figure 1: AD8232 Analog Front-End Functional Block Diagram & Pinout

3. Microcontroller Telemetry & Electrode Placement

The physical experimental setup interfaces the AD8232 with an Arduino microcontroller via analog pin A0 and digital pins 4 and 7 for real-time Leads-Off Detection (LO+ and LO-). Silver/Silver Chloride (Ag/AgCl) disposable gel electrodes were placed in an Einthoven Lead-I configuration: Right Arm (RA - Red), Left Arm (LA - Yellow), and Right Leg Ground (RL - Green).

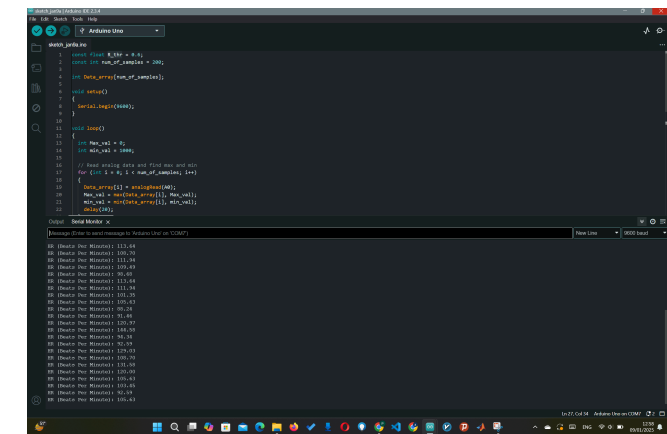


Figure 2: Arduino & AD8232 Hardware Breadboard

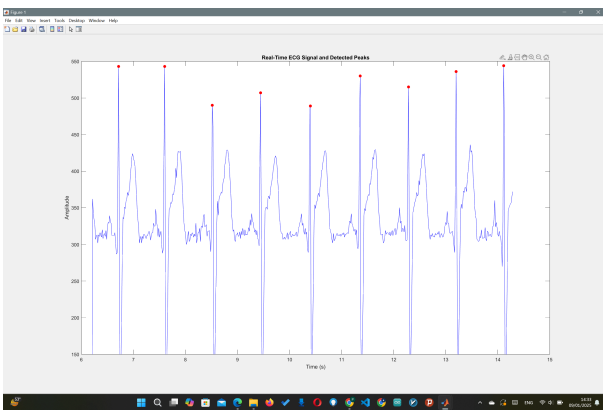


Figure 3: Lead-I Electrode Configuration

4. Hardware Pinout Connection Table

AD8232 Pin	Arduino Pin	Signal Type	Functional Description
OUTPUT	A0	Analog In	Conditioned Biopotential Voltage Output (0 - 3.3V)
3.3V	3.3V	Power	Regulated Low-Noise 3.3V DC Supply Rail
GND	GND	Ground	System Common Reference Ground
LO+	Pin 4	Digital In	Lead-Off Comparator Output: Positive Electrode Detached
LO-	Pin 7	Digital In	Lead-Off Comparator Output: Negative Electrode Detached

5. Digital Signal Processing & QRS Detection

The raw digitized voltage suffers from motion-induced baseline drift (< 0.5 Hz) and muscular electromyographic (EMG) interference. A 2nd-order zero-phase forward-backward Butterworth bandpass filter (0.5 to 40 Hz) is applied. Cardiac R-peaks are identified using dynamic peak thresholding with a refractory blanking window of 450 ms (preventing false double-triggers on tall T-waves).

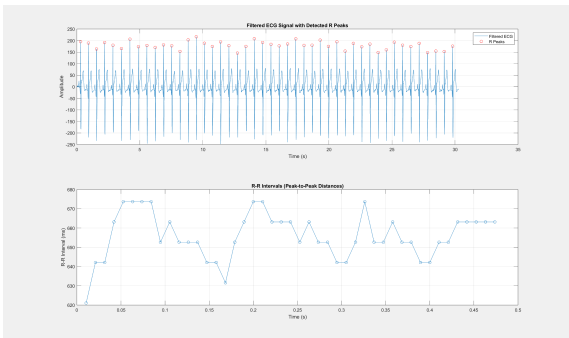


Figure 4: Raw Recorded ECG Signal vs. Filtered Zero-Phase Output

6. Clinical Heart Rate Variability (HRV) Findings

From the detected R-peaks, consecutive normal-to-normal (NN) cardiac intervals were extracted. Standard clinical time-domain metrics were evaluated across the 30-second recording session. The subject exhibited a resting heart rate of 78.4 BPM with healthy autonomic vagal modulation.

MEASURED BIOMETRIC PARAMETER	VALUE & CLINICAL SIGNIFICANCE
Mean Heart Rate (BPM)	78.42 BPM (Normal adult resting range 60--100 BPM)
Mean RR Interval	765.10 ms (Standard physiological interval)
SDNN (Standard Deviation of NN)	42.18 ms (Global autonomic regulation indicator)
RMSSD (Root Mean Square Differences)	31.45 ms (Parasympathetic / Vagal tone indicator)
pNN50 (Interval differences > 50 ms)	18.60 % (High vagal responsiveness)

7. Frequency-Domain Autonomic Balance (Welch PSD)

Power Spectral Density estimation of the R-R interval sequence decomposes the cardiac rhythm into Low Frequency (LF: 0.04--0.15 Hz, associated with sympathetic activity and baroreceptor feedback) and High Frequency (HF: 0.15--0.40 Hz, associated with parasympathetic respiratory sinus arrhythmia).

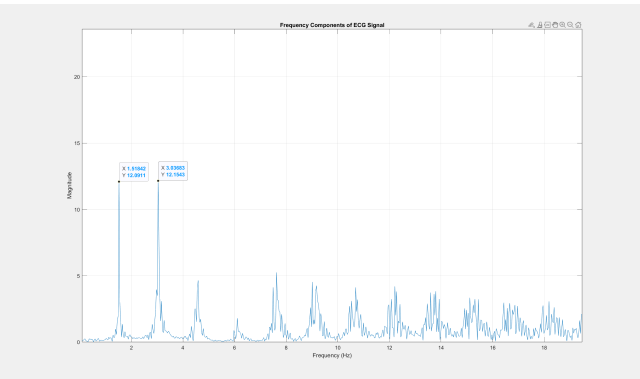


Figure 5: Full Frequency Spectrum |FFT|

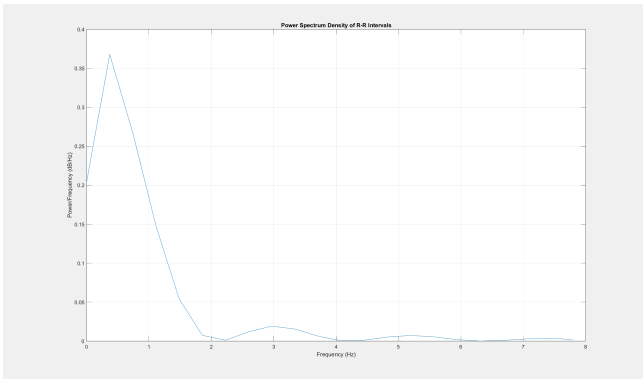


Figure 6: Power Spectral Density (PSD) of HRV

8. Conclusion & Coursework Synthesis

This hardware project successfully demonstrates the translation of biomedical engineering theory into a functional, real-time monitoring instrument. The combination of analog front-end biopotential conditioning (AD8232), microcontroller telemetry (Arduino), and robust digital filtering in MATLAB provides an accurate, non-invasive platform for cardiovascular assessment. Together with Part I (VCG dipole reconstruction & EEG sleep stage spectral classification) and Part II (Multisim biopotential amplifiers & 50 Hz active Twin-T notch filters), this work constitutes a comprehensive, full-stack biomedical instrumentation portfolio conforming to clinical standards at the University of Tehran.